## Experiment 8 -Practical Exercise: Set Up a Jenkins CI Pipeline for a Maven Project, Use Ansible to Deploy Artifacts Generated by Jenkins.

**Prerequisites:**

- Make sure Apache tomcat is installed in the root directory.
- Add the tomcat.service file if not available.

Set permissions for the username: tomcat using below commands,

**id tomcat**

**sudo chmod +x /opt/tomcat/bin/*.sh**

**sudo chown -R tomcat: tomcat /opt/tomcat**

Write tomcat.service file which is in /etc/systemd/system using below commands,

**sudo -i**

**cd /etc/systemd/system**

Edit the tomcat.service file using gedit and paste the following content.

**contents[Service]**

**Type=forking**

**User=Tomcat**

**Group=tomcat**

**Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64"**

**Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"**

**Environment="CATALINA_HOME=/opt/tomcat"**

**Environment="CATALINA_BASE=/opt/tomcat"**

**ExecStart=/opt/tomcat/bin/startup.sh**

**ExecStop=/opt/tomcat/bin/shutdown.sh**

**Restart=on-failure**

**[Install]**

**WantedBy=multi-user.target**

Create a maven-based web-application,

**mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false**

```
vboxuser@niranjan: ~/devops
vboxuser@niranjan:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenAnsibleApp -DarchetypeArtifactId=maven-archetype
-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/vboxuser/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenAnsibleApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/devops/MyMavenAnsibleApp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.870 s
[INFO] Finished at: 2026-05-10T22:21:44+05:30
[INFO] ------------------------------------------------------------------------
vboxuser@niranjan:~/devops$
```

Edit the pom.xml file by adding the java servelet dependency as given.

**<dependency>**

**<groupId>javax.servlet</groupId>**

**<artifactId>javax.servlet-api</artifactId>**

**<version>4.0.1</version>**
**<scope>provided</scope>**

**</dependency>**

File structure of the web-app will look like this,

```
vboxuser@niranjan: ~/devops/MyMavenAnsibleApp
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$ tree
.
├── pom.xml
└── src
    └── main
        ├── resources
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

5 directories, 3 files
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$
```

Create directory named ansible and navigate to it.

**mkdir ansible**

**cd ansible**

```
vboxuser@niranjan: ~/devops/MyMavenAnsibleApp/ansible
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$ mkdir ansible
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$ cd ansible/
vboxuser@niranjan:~/devops/MyMavenAnsibleApp/ansible$
```

Create hosts.ini file and open it using gedit, then add the given content. The hosts.ini (or inventory file) lists the IP addresses or hostnames of the target systems where Ansible will run the playbooks.

```
vboxuser@niranjan: ~/devops/MyMavenAnsibleApp/ansible
vboxuser@niranjan:~/devops/MyMavenAnsibleApp/ansible$ gedit hosts.ini
vboxuser@niranjan:~/devops/MyMavenAnsibleApp/ansible$ gedit playbook.yml
vboxuser@niranjan:~/devops/MyMavenAnsibleApp/ansible$
```

**[web]**

**Localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3**

Create playbook.yml file and the content as given below.

**Playbook:** A playbook.yml is an Ansible file written in YAML that defines a series of automation tasks to be executed on remote hosts.

After this, it's time to buid the maven web-app,

**mvn clean install**

```
vboxuser@niranjan: ~/devops/MyMavenAnsibleApp
vboxuser@niranjan:~/devops/MyMavenAnsibleApp/ansible$ cd ..
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyMavenAnsibleApp >--------------------
[INFO] Building MyMavenAnsibleApp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenAnsibleApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenAnsibleApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenAnsibleApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenAnsibleApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMavenAnsibleApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MyMavenAnsibleApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MyMavenAnsibleApp ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MyMavenAnsibleApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MyMavenAnsibleApp] in [/home/vboxuser/devops/MyMavenAnsibleApp/target/MyMavenAnsibleApp]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/vboxuser/devops/MyMavenAnsibleApp/src/main/webapp]
```

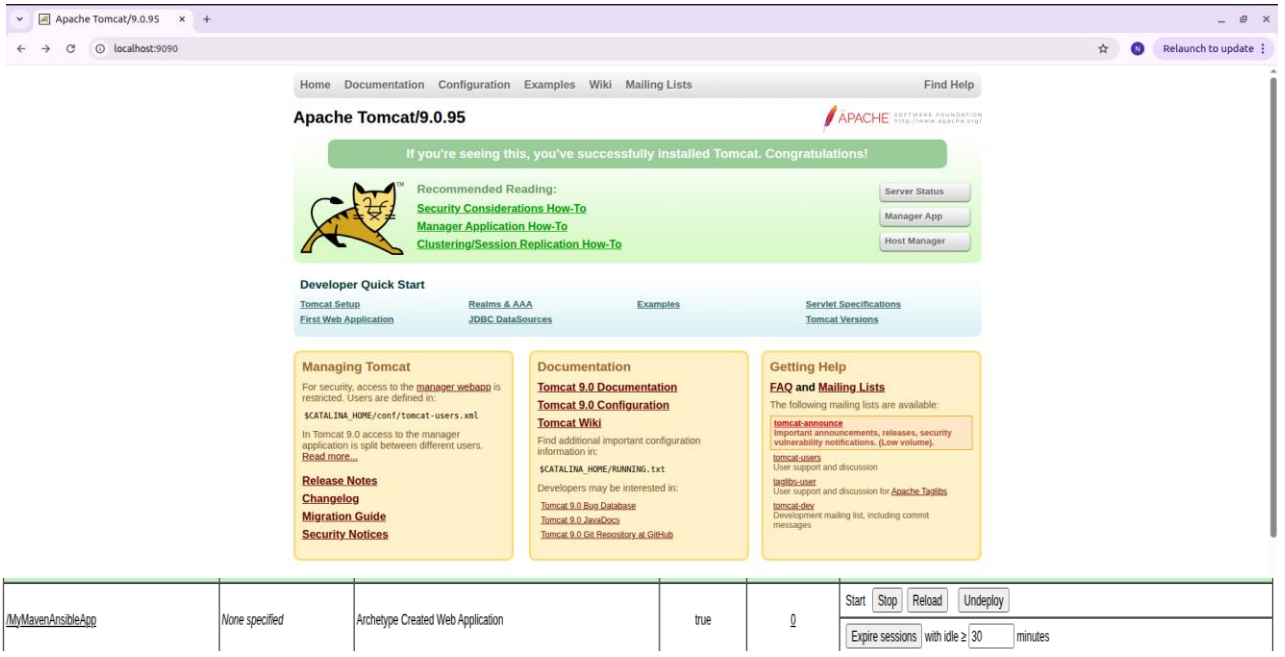
```
[INFO] Building war: /home/vboxuser/devops/MyMavenAnsibleApp/target/MyMavenAnsibleApp.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenAnsibleApp ---
[INFO] Installing /home/vboxuser/devops/MyMavenAnsibleApp/target/MyMavenAnsibleApp.war to /home/vboxuser/.m2/repository/com/example/MyMavenAn
sibleApp/1.0-SNAPSHOT/MyMavenAnsibleApp-1.0-SNAPSHOT.war
[INFO] Installing /home/vboxuser/devops/MyMavenAnsibleApp/pom.xml to /home/vboxuser/.m2/repository/com/example/MyMavenAnsibleApp/1.0-SNAPSHOT
/MyMavenAnsibleApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.189 s
[INFO] Finished at: 2026-05-10T22:37:05+05:30
[INFO] ------------------------------------------------------------------------
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$
```

Start the Tomcat Server as shown below:

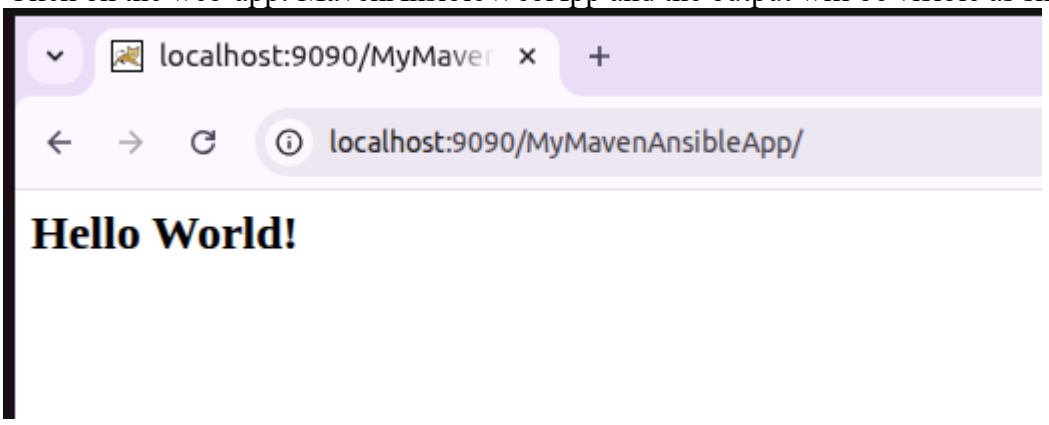
```
root@niranjan: /opt/tomcat/bin
vboxuser@niranjan:~/devops/MyMavenAnsibleApp$ cd ..
vboxuser@niranjan:~/devops$ cd ..
vboxuser@niranjan:~$ sudo -i
[sudo] password for vboxuser:
root@niranjan:~# cd /opt/tomcat/webapps
root@niranjan:/opt/tomcat/webapps# ls
docs      host-manager  MavenAnsibleWebApp1      mavenWeb      MyMavenWebApp      ROOT
examples  manager       MavenAnsibleWebApp1.war  mavenWeb.war  MyMavenWebApp.war
root@niranjan:/opt/tomcat/webapps# cd ..
root@niranjan:/opt/tomcat# cd bin
root@niranjan:/opt/tomcat/bin# ls
bootstrap.jar       ciphers.bat                  configtest.bat  digest.sh         setclasspath.sh  startup.sh            tool-wrapper.sh
catalina.bat        ciphers.sh                   configtest.sh   makebase.bat      shutdown.bat     tomcat-juli.jar       version.bat
catalina.sh         commons-daemon.jar           daemon.sh       makebase.sh       shutdown.sh      tomcat-native.tar.gz  version.sh
catalina-tasks.xml  commons-daemon-native.tar.gz  digest.bat      setclasspath.bat  startup.bat      tool-wrapper.bat
root@niranjan:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@niranjan:/opt/tomcat/bin#
```

Visit http://localhost:9090 and click on manager app and login with the user credentials,

i.e., user=admin, password=Admin




Click on the web-app: MavenAnsibleWebApp and the output will be visible as shown.



To create a Jenkins pipeline, create a Jenkins pipeline script named JenkinsFile with the below content.

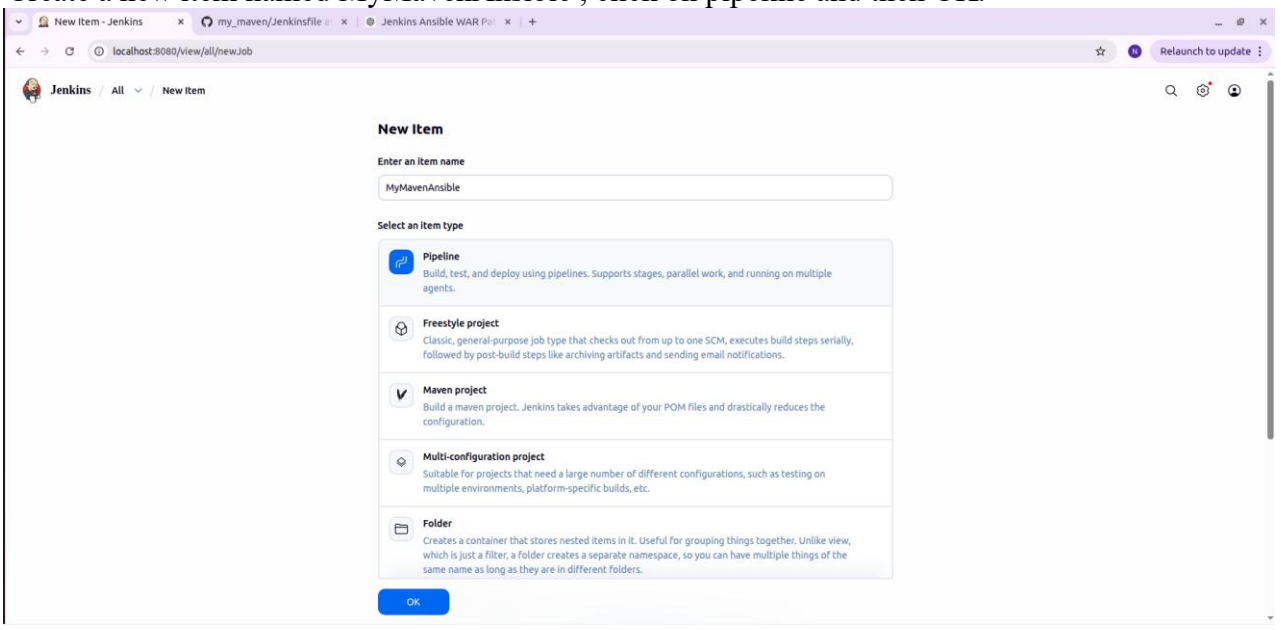
```
pipeline {
agent any  // Use any available agent
environment {
        LANG = 'en_US.UTF-8'
        LC_ALL = 'en_US.UTF-8'
}
tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
}
stages {
        stage('Checkout') {
                steps {
                        git url 'https://github.com/NiranjanGK/MyMavenAnsible.git
                }
        }
        stage('Build') {
                steps {
                        sh 'mvn clean package'
                }
        }
        stage('Archive') {
                steps {
                archiveArtifacts artifacts: 'target/*.war', fingerprint:true
                }
        }
```

```
        stage('Deploy') {
                steps {
                        sh 'mvn clean package'
                        sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
                }
        }
}
}
```

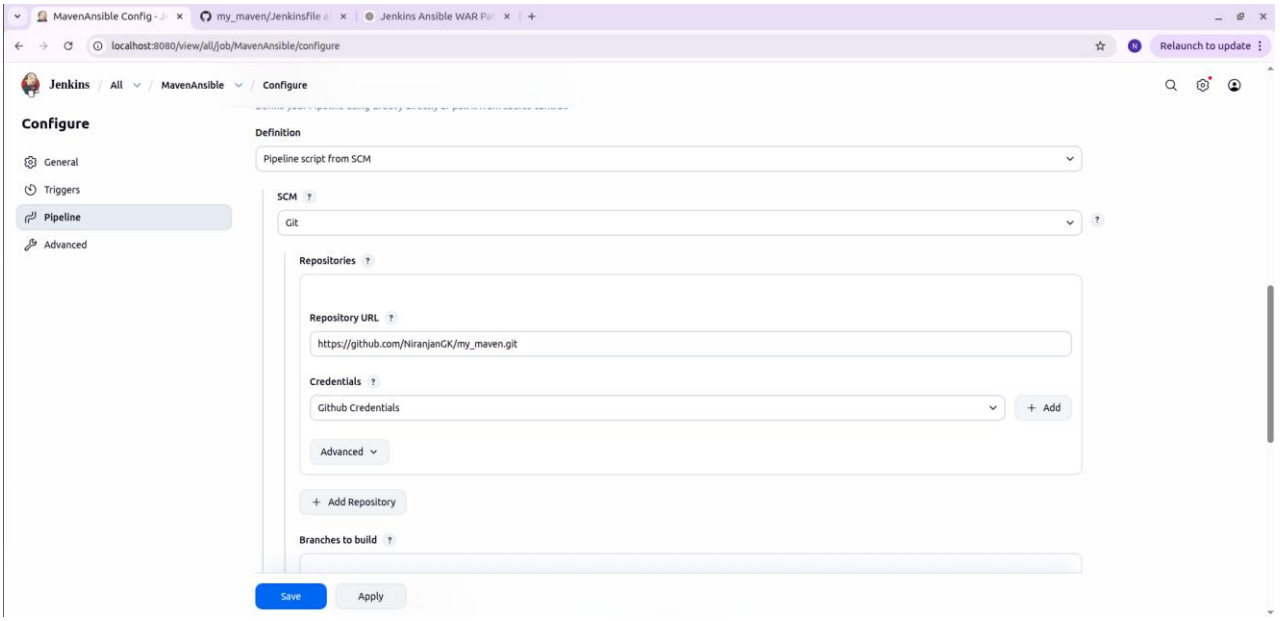
Push the complete webapp into github.

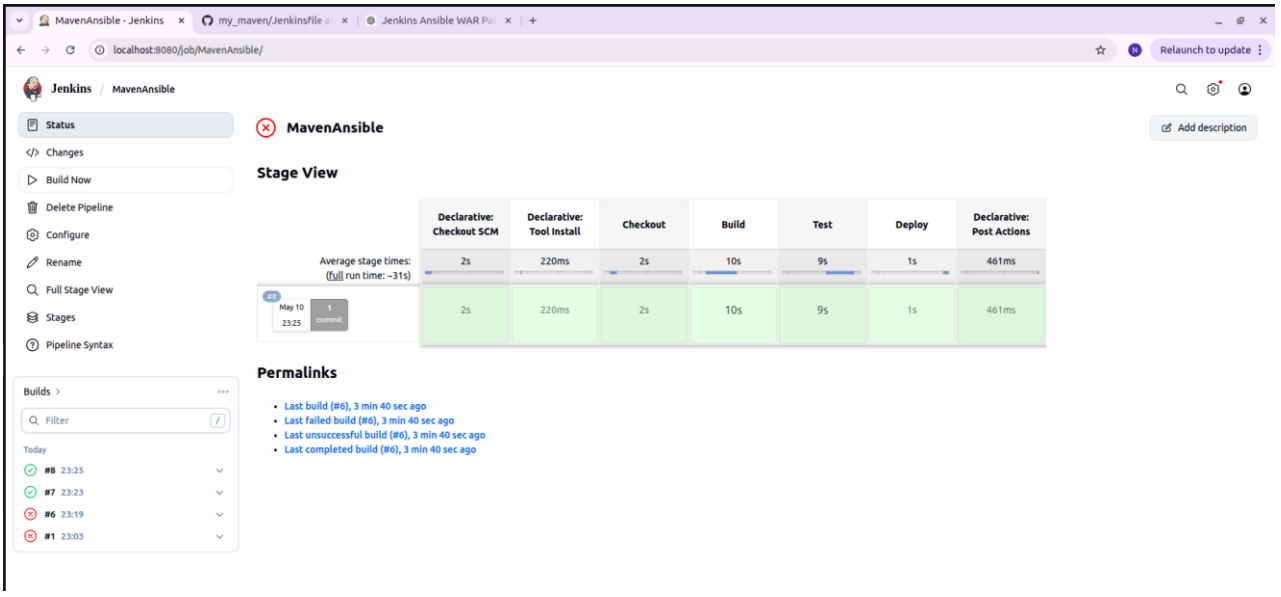
Open Jenkins by visiting http://localhost:8080.

Create a new item named MyMavenAnsible , click on pipeline and then OK.




Edit configuration to extract Jenkins pipeline script from github.




Click on Build Now